

The big idea. Two or more quantities change over time t , and one equation ties them together. Differentiate that equation with respect to t , and you get an equation that ties their *rates* together. Know one rate, solve for the other.

THE SIX STEPS

- 1

Draw and label. Sketch the situation. Give every *changing* quantity a variable. Label anything that never changes with its number.
- 2

Given and find. Write each known rate as a derivative with units, e.g. $\frac{dV}{dt} = 100 \text{ cm}^3/\text{s}$. Shrinking means **negative**. Circle the rate you want.
- 3

Relate the variables. Write one equation linking them (see the formula bank). If it has an extra variable, eliminate it first, usually with similar triangles.
- 4

Differentiate with respect to t . Every variable gets the chain rule: $\frac{d}{dt}[r^3] = 3r^2 \frac{dr}{dt}$. Constants become 0.
- 5

Substitute the snapshot. Now plug in the values at the given instant. Use the Step 3 equation to find any value you are missing.
- 6

Solve and interpret. Solve for the unknown rate. Answer in a sentence with units. The sign tells you increasing (+) or decreasing (-).

Most common mistake: plugging in snapshot numbers *before* Step 4. Plug in $r = 5$ first and r becomes a constant with derivative 0. **Differentiate first, then substitute.**

FORMULA BANK

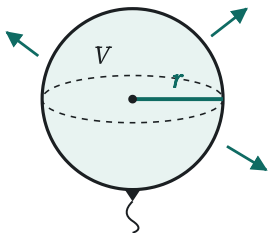
CIRCLES & SPHERES	CYLINDERS & CONES	BOXES & FLAT SHAPES
Circle area $A = \pi r^2$	Cylinder volume $V = \pi r^2 h$	Cube $V = s^3, S = 6s^2$
Circumference $C = 2\pi r$	Cylinder surface $S = 2\pi r^2 + 2\pi rh$	Box $V = lwh$
Sphere volume $V = \frac{4}{3}\pi r^3$	Cone volume $V = \frac{1}{3}\pi r^2 h$	Rectangle $A = lw, P = 2l + 2w$
Sphere surface $S = 4\pi r^2$	Cone-shaped tank $\frac{r}{h} = \frac{R}{H}$ (similar $\triangle$ s)	Triangle $A = \frac{1}{2}bh$

RIGHT TRIANGLES	OTHER TRIANGLES	DIFFERENTIATING IN T
Pythagorean $x^2 + y^2 = z^2$	Similar $\triangle$ s $\frac{\text{height}_1}{\text{base}_1} = \frac{\text{height}_2}{\text{base}_2}$	Power $\frac{d}{dt}[x^n] = nx^{n-1} \frac{dx}{dt}$
Sine, cosine $\sin \theta = \frac{\text{opp}}{\text{hyp}}, \cos \theta = \frac{\text{adj}}{\text{hyp}}$	Law of cosines $c^2 = a^2 + b^2 - 2ab \cos \theta$	Product $\frac{d}{dt}[xy] = x \frac{dy}{dt} + y \frac{dx}{dt}$
Tangent $\tan \theta = \frac{\text{opp}}{\text{adj}}$	Distance $D = \sqrt{(\Delta x)^2 + (\Delta y)^2}$	Sine, cosine $\cos \theta \frac{d\theta}{dt}, -\sin \theta \frac{d\theta}{dt}$
		Tangent $\frac{d}{dt}[\tan \theta] = \sec^2 \theta \frac{d\theta}{dt}$

Walkthrough 1: The inflating balloon

WARM-UP LEVEL

Air is pumped into a spherical balloon at $100 \text{ cm}^3/\text{s}$. How fast is the radius increasing at the instant the radius is 5 cm?



1 DRAW

r = radius and V = volume. Both change as air goes in.

2 GIVEN

$\frac{dV}{dt} = 100 \text{ cm}^3/\text{s}$, snapshot $r = 5 \text{ cm}$. **Find:** $\frac{dr}{dt}$.

3 RELATE

$V = \frac{4}{3}\pi r^3$

4 DIFF.

$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$ (power rule, then chain rule adds $\frac{dr}{dt}$)

5 PLUG IN

$100 = 4\pi(5)^2 \frac{dr}{dt} = 100\pi \frac{dr}{dt}$

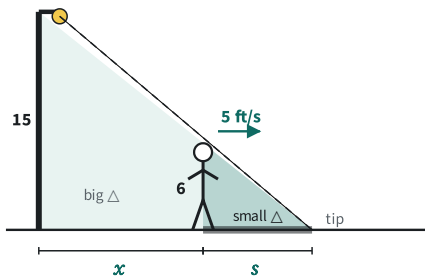
6 SOLVE

$\frac{dr}{dt} = \frac{1}{\pi} \approx \mathbf{0.318 \text{ cm/s}}$ The radius grows about 0.32 cm each second.

Walkthrough 2: The shadow

SIMILAR TRIANGLES

A 6-ft-tall man walks away from a 15-ft streetlight at 5 ft/s. When he is 40 ft from the pole, (a) how fast is his shadow getting longer? (b) How fast is the tip of his shadow moving?



1 DRAW x = man-to-pole distance, s = shadow length. The 15 and 6 never change.

2 GIVEN $\frac{dx}{dt} = 5$ ft/s, snapshot $x = 40$ ft. **Find:** (a) $\frac{ds}{dt}$, (b) rate of the tip.

3 RELATE The two shaded triangles are similar (same light ray):

$$\frac{15}{x+s} = \frac{6}{s} \Rightarrow 15s = 6x + 6s \Rightarrow s = \frac{2}{3}x$$

4 DIFF. $\frac{ds}{dt} = \frac{2}{3} \frac{dx}{dt}$

5 PLUG IN $\frac{ds}{dt} = \frac{2}{3}(5)$

6 SOLVE (a) $\frac{ds}{dt} = \frac{10}{3} \approx \boxed{3.33 \text{ ft/s}}$ Shadow grows 3.33 ft each second.

(b) The tip. A new quantity needs a new variable.

3 RELATE Tip-to-pole distance: $y = x + s = x + \frac{2}{3}x = \frac{5}{3}x$

4-5 $\frac{dy}{dt} = \frac{5}{3} \frac{dx}{dt} = \frac{5}{3}(5)$

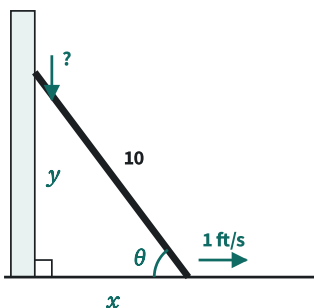
6 SOLVE $\frac{dy}{dt} = \frac{25}{3} \approx \boxed{8.33 \text{ ft/s}}$ This is the man's 5 ft/s plus the shadow's growth of $\frac{10}{3}$ ft/s.

Notice: we never used $x = 40$. Since $s = \frac{2}{3}x$ is linear, the rate is the same at every distance. Not every number has to be used.

Walkthrough 3: The sliding ladder

PYTHAGOREAN + TRIG

A 10-ft ladder leans against a wall. The bottom slides away from the wall at 1 ft/s. When the bottom is 6 ft from the wall, (a) how fast is the top sliding down? (b) How fast is the angle between the ladder and the ground changing?



1 DRAW x = bottom-to-wall, y = height of top, θ = angle with ground. The 10 never changes.

2 GIVEN $\frac{dx}{dt} = 1$ ft/s, snapshot $x = 6$ ft. **Find:** (a) $\frac{dy}{dt}$, (b) $\frac{d\theta}{dt}$.

3 RELATE $x^2 + y^2 = 10^2$

4 DIFF. $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$ (10^2 is a constant, so its derivative is 0)

5 PLUG IN Missing y ? Use Step 3: $6^2 + y^2 = 100 \Rightarrow y = 8$. Then $2(6)(1) + 2(8) \frac{dy}{dt} = 0$.

6 SOLVE (a) $\frac{dy}{dt} = -\frac{12}{16} = \boxed{-0.75 \text{ ft/s}}$ Negative: the top slides down.

(b) The angle. Pick the trig ratio that uses the known rate (x) and a constant (10).

3 RELATE $\cos \theta = \frac{x}{10}$

4 DIFF. $-\sin \theta \frac{d\theta}{dt} = \frac{1}{10} \frac{dx}{dt}$

5 PLUG IN $\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{8}{10}$; $-\frac{8}{10} \frac{d\theta}{dt} = \frac{1}{10}(1)$

6 SOLVE (b) $\frac{d\theta}{dt} = -\frac{1}{8} = \boxed{-0.125 \text{ rad/s}}$ The angle is shrinking.